

Local connectivity of Julia sets for rational maps with Siegel disks

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MINI WORKSHOP ON DYNAMICS

Wuhan University

Feb 18, 2023

Complex dynamical systems

$$f : X \rightarrow X$$

$$x \mapsto f(x) \mapsto f(f(x)) \mapsto f(f(f(x))) \mapsto \cdots \mapsto f^{\circ n}(x) \mapsto \cdots$$

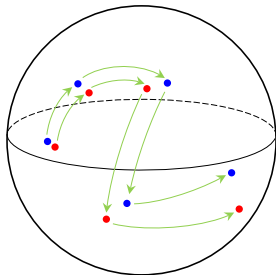
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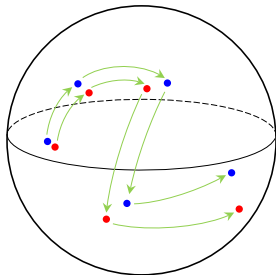
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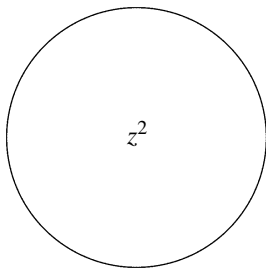
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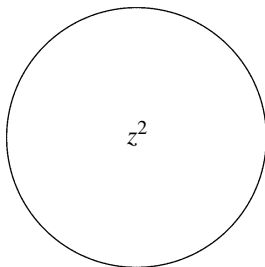
The **Julia set** (or chaotic set) $J(f) := \widehat{\mathbb{C}} \setminus F(f)$.

Each connected component of $F(f)$ (resp. $J(f)$) is called a **Fatou** (resp. **Julia**) **component**.

Julia / Fatou sets

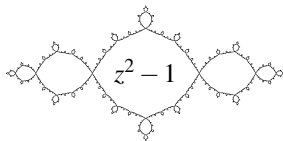
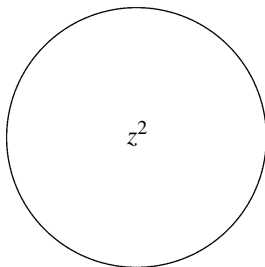


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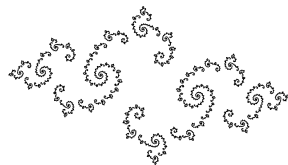
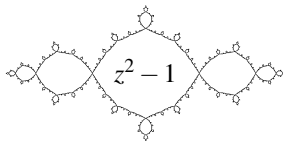
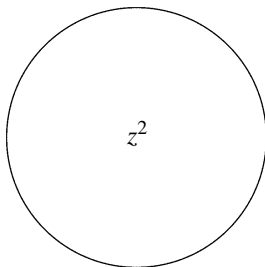


$$z^2 - 1?$$

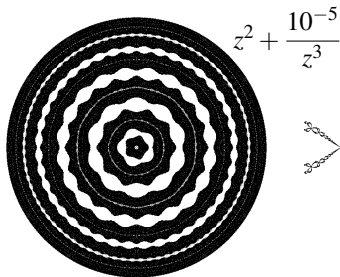
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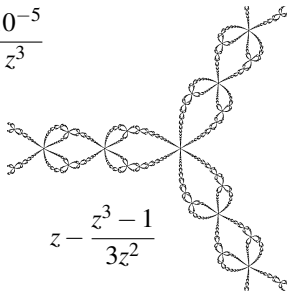
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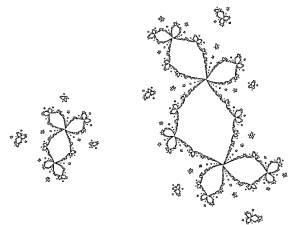
$$z^2 - 0.78 + 0.23i$$



$$z^2 + \frac{10^{-5}}{z^3}$$



$$z - \frac{z^3 - 1}{3z^2}$$



$$z^3 - 0.48z + 0.7 + 0.5i$$

Sullivan's eventually periodic theorem

Theorem (Sullivan, Ann. Math., 1985)

The Fatou components of all rational maps are eventually periodic.

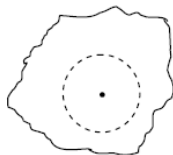
Classification of periodic Fatou components of rational maps:



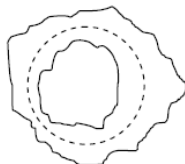
(super-)attracting basin
1884, 1904



parabolic basin
1897

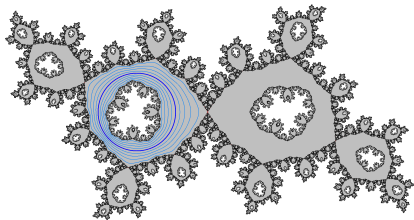
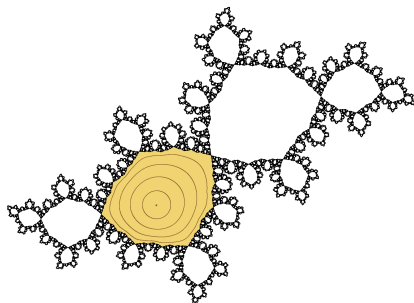
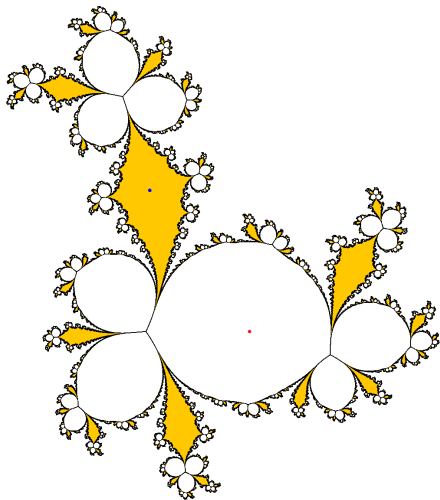


Siegel disk
1942



Herman ring
1979

Periodic Fatou components



Critical points

Critical point c : $f'(c) = 0$, with local expansion

$$f(z) = c + a(z - c)^k + \cdots, \quad \text{where } k \geq 2.$$

Critical points and their orbits play an essential role in the dynamics of rational maps (**contraction** vs **expanding**).

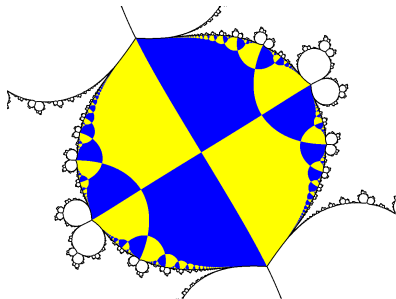
Fatou (1920s):

- Every **attracting** or **parabolic** basin contains at least one critical point.
- Every point in the boundary of **Siegel** disk or **Herman** ring is accumulated by the orbit of at least one critical point.

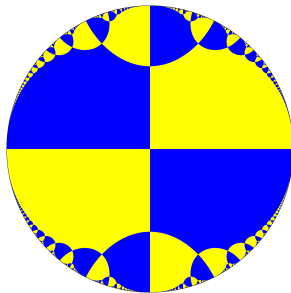
Dynamics in periodic Fatou components

Lemma (McMullen, 1988)

A rational map with a simply connected periodic Fatou component is conjugated there to a Blaschke product on $\mathbb{D}$ with $J = \partial\mathbb{D}$.



A parabolic basin of $e^{2\pi i/3}z + z^2$



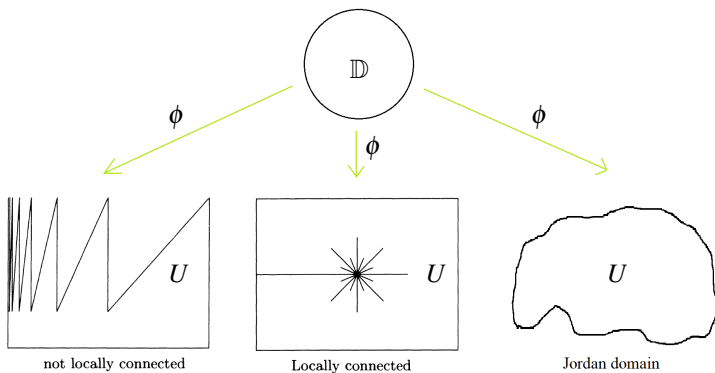
Blaschke model $\frac{z^d + a_d}{1 + a_d z^d}$, $a_d = \frac{d-1}{d+1}$, $d = 2$

Study Julia sets from Fatou sets

Carathéodory (1913):

The biholomorphic map $\phi : \mathbb{D} \rightarrow U \subset \widehat{\mathbb{C}}$ can be extended to a continuous map $\bar{\phi} : \overline{\mathbb{D}} \rightarrow \overline{U} \iff \partial U$ is **Locally Connected**.

In particular, $\bar{\phi}$ is a homeomorphism $\iff \partial U$ is a Jordan curve.

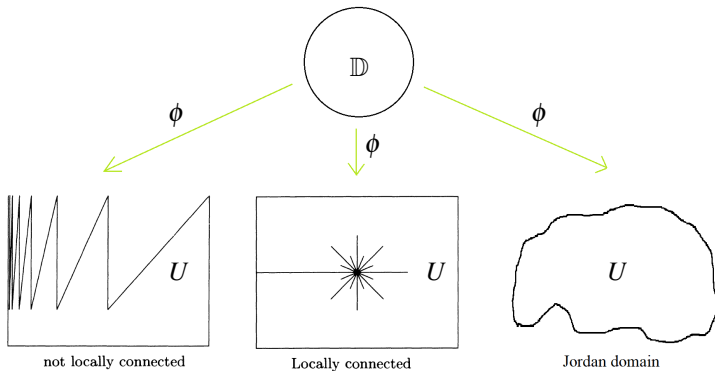


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If LC holds: Dynamics from Fatou components to the Julia set continuously.

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Expanding metrics: Hyperbolic, orbifold, ...

Combinatorial tools: Puzzles developed by Yoccoz, Branner-Hubbard, Lyubich, Kozlovski-Shen-van Strien, Roesch, ...

Analytic tools: Grötzsch inequality on modulus, Kahn-Lyubich covering lemma ...

Siegel disk and continued fractions

$f(z) = e^{2\pi i \alpha} z + \mathcal{O}(z^2)$, where $\alpha \in \mathbb{R} \setminus \mathbb{Q}$.

Siegel disk Δ_f : The *maximal* region in which f is conjugate to $\zeta \mapsto e^{2\pi i \alpha} \zeta$.

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Let

$$\alpha = [0; a_1, a_2, \dots, a_n, \dots] = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots}}}$$

be the **continued fraction expansion**. Then $\frac{p_n}{q_n} = [0; a_1, a_2, \dots, a_n] \rightarrow \alpha$ exp. fast.

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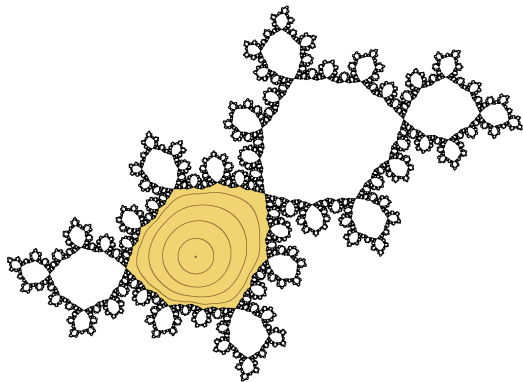
Theorem (Siegel, Ann. Math., 1942)

The holomorphic germ f has a Siegel disk at 0 if $\alpha \in \bigcup_{\kappa \geq 2} \mathcal{D}_\kappa$, where

$$\mathcal{D}_\kappa = \left\{ \alpha : \sup_{n \geq 1} \left\{ \frac{a_{n+1}}{q_n^{\kappa-2}} \right\} < \infty \right\}.$$

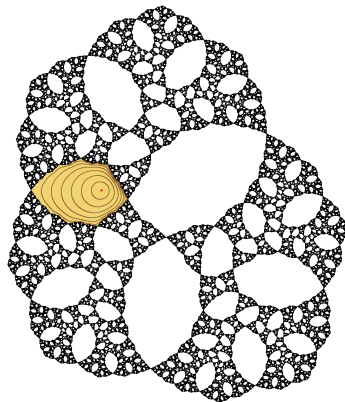
Remark: α is called **bounded type** if $\alpha \in \mathcal{D}_2$, i.e. $\sup_n \{a_n\} < \infty$.

Siegel disks



The Siegel disk of $f(z) = e^{2\pi i \alpha} z + z^2$, where

$$\alpha = \frac{\sqrt{5}-1}{2} = [0; 1, 1, 1, \dots]$$



The Siegel disk of $f(z) = \frac{e^{2\pi i \alpha} z}{(1-z)^2}$

Douady-Sullivan's conjecture

Conjecture (Douady-Sullivan, 1986)

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When α is of **bounded type**:

Theorem (Zhang, Invent. Math., 2011)

The bounded type Siegel disk of any rational map ($\deg \geq 2$) is a quasi-disk.

- (Douady-Ghys-Herman-Świątek, 1987) quadratic poly
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Theorem (Zakeri, Duke Math. J., 2010)

The bounded type Siegel disk of any non-linear $f(z) = P(z)e^{Q(z)}$ is a quasi-disk, where P, Q are polys., $f(0) = 0, f'(0) = \lambda = e^{2\pi i \alpha}$.

- (Geyer, 2001) $f(z) = \lambda z e^z$
- (Keen-Zhang, 2009)
 $f(z) = (\lambda z + az^2)e^z$

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- (Zhang, 2005) $f(z) = \lambda \sin(z)$
- (Y., 2013) $f(z) = \lambda \sin(z) + a \sin^3(z)$
- (Zhang-Fu-Shi, 2020) some $P(\sin(z))$ with P poly.
- (Chéritat, 2006) some “simple” entire functions
- (Chéritat-Epstein, 2018) some holo. maps with at most 3 singular values.

LC of Julia sets: with Siegel disks

Theorem (Petersen, Acta Math., 1996)

For any bounded type α , the Julia set of $P(z) = e^{2\pi i \alpha} z + z^2$ is locally connected.

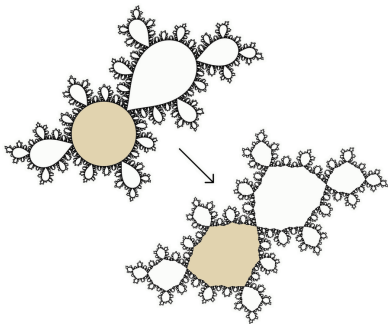
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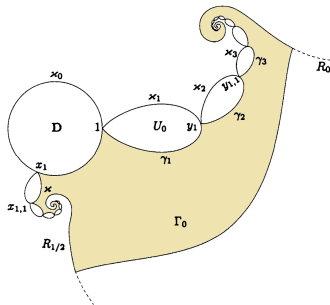
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$$\text{Blaschke model } f_\alpha(z) = e^{2\pi i t(\alpha)} z^2 \frac{z-3}{1-3z}$$



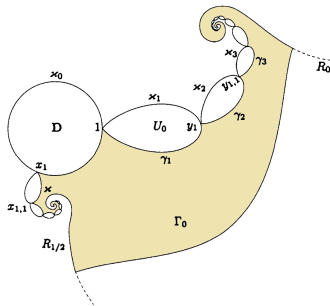
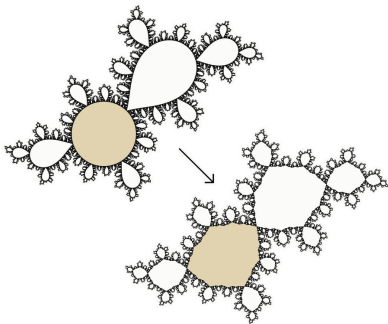
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$$\text{Blaschke model } f_\alpha(z) = e^{2\pi i t(\alpha)} z^2 \frac{z-3}{1-3z}$$

Remark: $J(P)$ is LC for **almost all** α : $\log a_n = \mathcal{O}(\sqrt{n})$ (Petersen-Zakeri, Ann. Math., 2004).

Complex a priori bounds

Based on **Petersen puzzles** and **complex a priori bound** for **unicritical** circle maps:

- Alternative proof of Petersen's result ([Yampolsky](#), 1999);
- Mating two quadratic Siegel disks ([Yampolsky-Zakeri](#), 2001);
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[Theorem \(J. Yang, arXiv, 2021\)](#)

*The Julia set of any **polynomial** (assumed to be connected) is locally connected at the boundary points of their bounded type Siegel disks.*

Proof tools: Petersen-like puzzle pieces + Kahn-Lyubich's covering lemma + complex a prior bound for **multicritical** circle maps.

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Our aim: Study the local connectivity of the Julia sets of **general rational maps** with bounded type Siegel disks.

Main result

Main Theorem (Wang-Y.-Zhang-Zhang, arXiv, 2022)

Suppose f is a **rational** map with Siegel disks such that $J(f)$ is connected, and moreover, the forward orbit of every critical point of f satisfies one of the following:

- ① It is finite; or
- ② It lies in an attracting basin; or
- ③ It intersects the closure of a bounded type Siegel disk.

Then $J(f)$ is locally connected.

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Main result

Main Theorem (Wang-Y.-Zhang-Zhang, arXiv, 2022)

Suppose f is a **rational** map with Siegel disks such that $J(f)$ is connected, and moreover, the forward orbit of every critical point of f satisfies one of the following:

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It provides an alternative proof of Petersen's result without using puzzles.

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Immediate consequences: The Julia sets of following maps are locally connected:

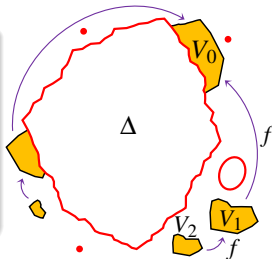
- Cubic polynomials in Zakeri's curves;
- Cubic Newton maps with a bounded type Siegel disk;
- McMullen maps $z \mapsto z^m + \lambda/z^n$ ($m \geq 2, n \geq 1$) with a bounded type Siegel disk.

Main Lemma

The postcritical set of f is $\mathcal{P}(f) := \overline{\bigcup_{n \geq 1} f^n(\text{Crit}(f))}$.

Main Lemma (WYZZ, 2022)

Let f be $\deg \geq 2$ rational having a fixed bounded type Siegel disk Δ with $\text{dist}_{\widehat{\mathbb{C}}}(\mathcal{P}(f) \setminus \partial\Delta, \partial\Delta) > 0$. Then $\forall \varepsilon > 0$, and $\forall$ Jordan domain $V_0 \subset \widehat{\mathbb{C}} \setminus \overline{\Delta}$ with $\emptyset \neq \overline{V_0} \cap \mathcal{P}(f) \subset \partial\Delta$, $\exists N = N(\varepsilon, V_0, f) \geq 1$, s.t. $\text{diam}_{\widehat{\mathbb{C}}}(V_n) < \varepsilon$ for all $n \geq N$, where V_n is any connected component of $f^{-n}(V_0)$.

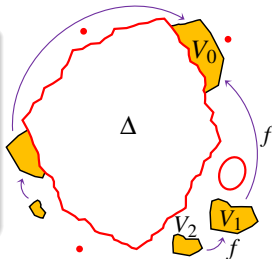


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Parallel to **classical Shrinking Lemma** (Tan-Yin 1996, Lyubich-Minsky 1997):

$$\overline{V_0} \cap \mathcal{P}(f) = \emptyset.$$

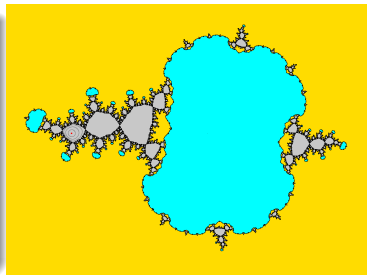
Consequence (Zhang et al): Bounded type Siegel and subhyperbolic quadratic polynomials with connected Julia sets are **mateable**.

Mechanism of the proof

Lemma (Torhost-Whyburn, 1942)

A compact set X in $\widehat{\mathbb{C}}$ is LC $\iff X$ satisfies:

- The boundary of each component of $\widehat{\mathbb{C}} \setminus X$ is LC; and
- $\forall \varepsilon > 0$, there are only finitely many components of $\widehat{\mathbb{C}} \setminus X$ whose spherical diameters are greater than ε .

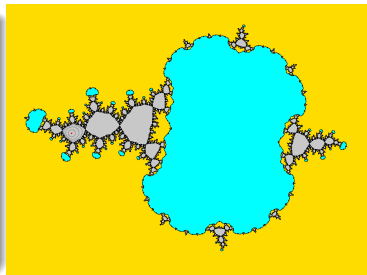


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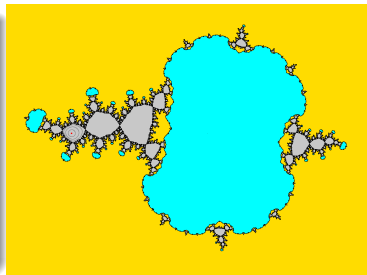
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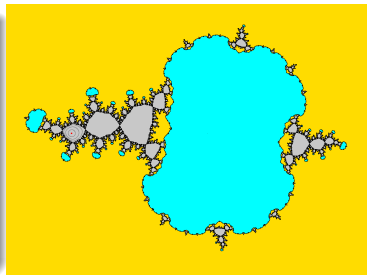
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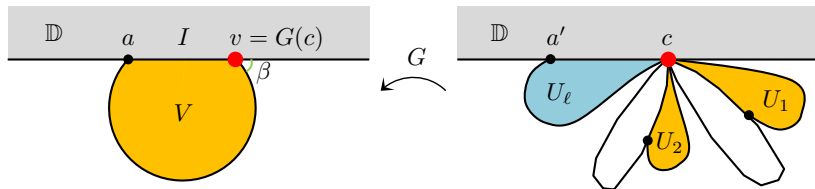
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Proof of the Main Theorem: (1) Zhang + (2) Proposition +
(3) Control sizes of Fatou components (Main Lemma + Classical Shrinking Lemma).

Quasi-Blaschke model and contraction

G : Quasi-Blaschke product model; $\mathcal{P}(G)$: The “postcritical set” of G .

$\Omega :=$ The unique component of $\widehat{\mathbb{C}} \setminus (\mathcal{P}(G) \cup \overline{\mathbb{D}})$ with $\mathbb{T} \subset \partial\Omega$.



$\exists \mu > 1$ depending only on V (only on β if $V = H^\beta(I)$) s.t.

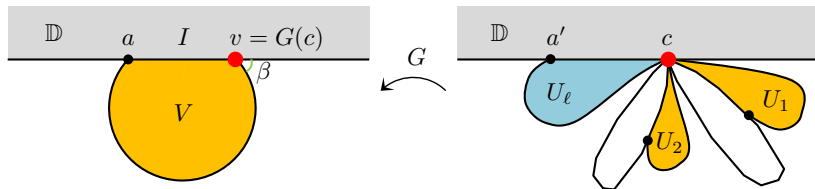
$$\|DG(z)\|_{\rho_\Omega} > \mu,$$

for all $z \in \bigcup_{1 \leq i \leq \ell-1} U_i$, where G has local degree $2\ell - 1$ at c with $\ell \geq 2$.

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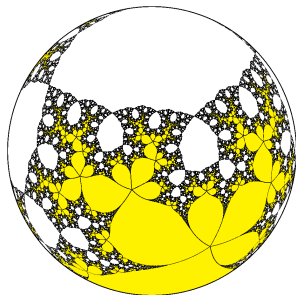
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Principle to prove the Main Lemma: Pulling back $V_0 = H^\beta(I)$, we have $|\overline{V}_n \cap \mathbb{T}| \rightarrow 0$ ($n \rightarrow \infty$) and apply the contraction successively.

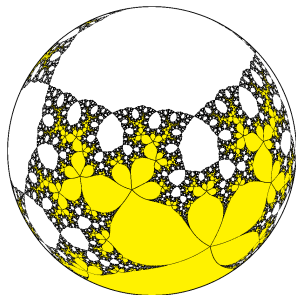
Further developments



Work in progress (Fu-Y., based on the Main Lemma):

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Potential topics and challenges:

- Extending the bounded type rotation numbers to unbounded type;
- Allowing the critical orbits to have larger degrees of freedom.

Thank you for your attention!